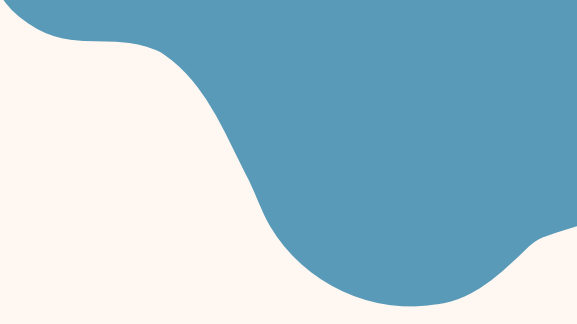




PYTHON – WORKING WITH CSV FILES



TODAY'S LEARNING JOURNEY

WHAT IS CSV?

Learn that CSV is like a digital table that stores data in rows and columns!

MEMORY VS STORAGE

Understand why variables forget data but CSV files remember forever!

READING CSV FILES

Use Python's csv module to read attendance, IPL points, and grocery lists!



THE PROBLEM

Variables are like chalk on a blackboard. When you restart your program, everything gets wiped clean! All your data disappears like magic erased from the board.



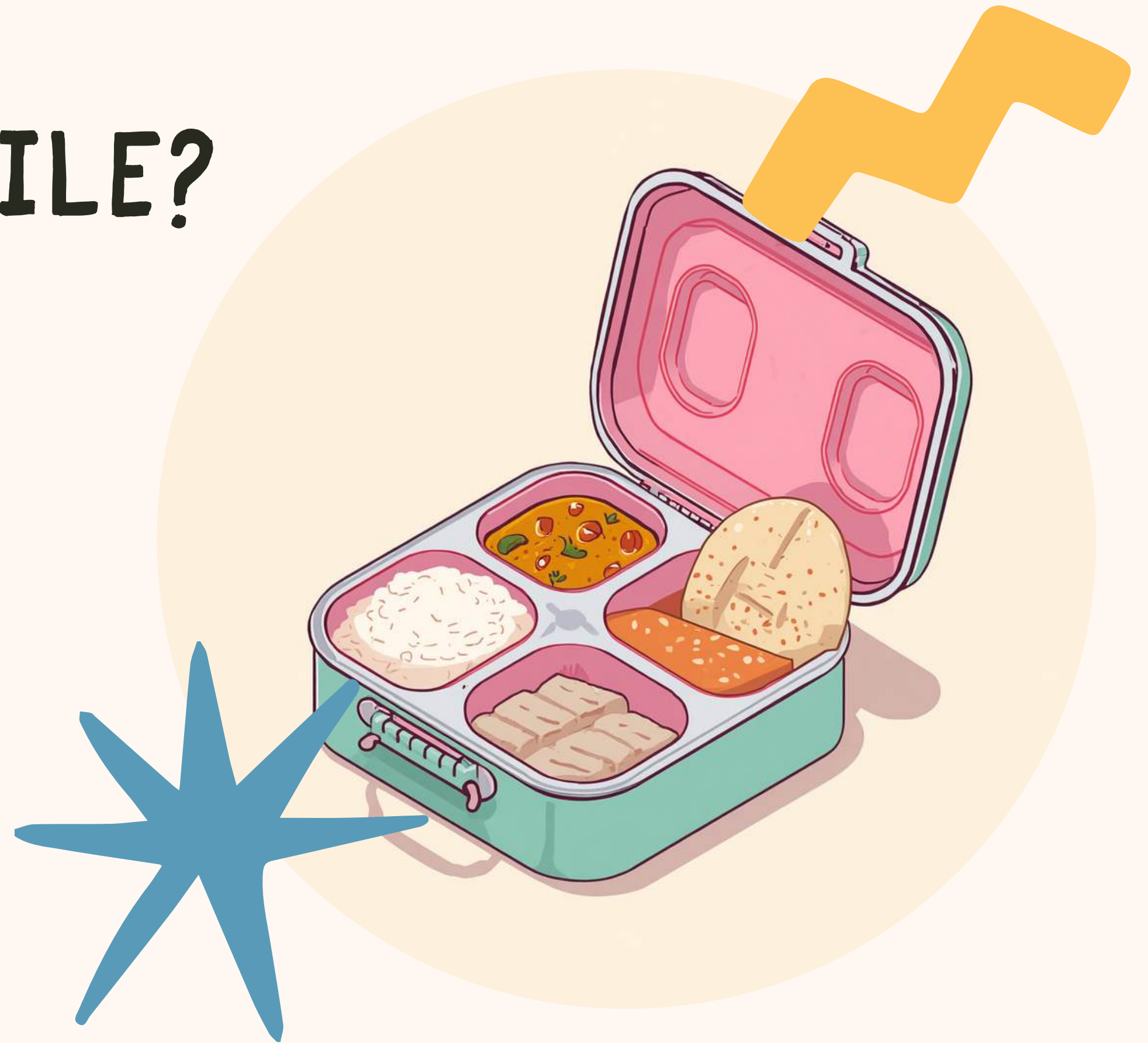
THE SOLUTION

Permanent Storage is like writing in a notebook. Your data stays safe forever, even after the program ends. CSV files remember everything!

WHAT IS A CSV FILE?

CSV stands for "Comma Separated Values" - it's like a digital tiffin box! Just like your tiffin has compartments separated by walls, a CSV file has data separated by commas. Each comma creates a new compartment for your information!

Example: Rahul, 10, Mumbai becomes three neat compartments of data!



ROWS AND COLUMNS IN CSV



Think of a CSV file like your school timetable! Rows are horizontal lines going across – like each day of the week. Columns are vertical sections going down – like each period. Together, they create a neat table where every piece of data has its own special spot!



School Timetable

SUNDAY	FRIDAY	THUESDAY	SEDINDAY	TUTURD.
Thursday	Jochar 12	Bodker 14	Jughar 15	12:12 Octobes 10:7
Hindi				
Sciemer 16	Jchorr 18	Bodker 19	Jafner 18	12:10 October: 2:13
Scilsbec				
Tumber 15	Acher 29	Sedngr 18	Juler 28	12:10 Cctoes 10:2
ience 14				
	Acion 28	Scdngr 21	Julat 29	

SEEING INSIDE A CSV FILE

Raw CSV in Notepad:

Name,Marks,Grade

Aarav,85,A

Priya,72,B

Rohan,91,A

See the commas? They separate each piece of data!



Student Dates

Name	Name	Priya	Grade	Grade	Privid
Aaraiv	16	39	10	1	21
Rerarie	19	16	19	4	24
Aumm	36	26	38	8	27
Rarain	26	20	26	3	22
Raram	39	20	29	12	33
Pando	16				
Demme	16	10	14	4	15
Sarao	13	19	10	5	10
Miyan	10	16	16	9	10
Saned	11				
Rohan	16	30	14	1	10
Raban	16	40	13	8	23
Rohan	16	40	14	10	10

SPREADSHEET VIEW

Same file in Excel or Google

Sheets shows neat rows and

columns – much easier to read!

INDIAN DATASET EXAMPLES



ATTENDANCE REGISTER

Track who comes to class!
Columns: Student Name, Roll
Number, Present (Yes/No).
Easy to mark daily attendance.



IPL POINTS TABLE

Follow your favorite cricket
teams! Columns: Team Name,
Matches Won, Total Points. Just
like on TV!



GROCERY LIST

Help at home with shopping!
Columns: Item Name, Quantity,
Price. Perfect for tracking
Vegetable and dal!



READING CSV FILES IN PYTHON

THE CSV MODULE

Python has a built-in 'csv' module that helps us read CSV files easily. We use 'with open' to safely open files!

SAMPLE CODE TO READ CSV

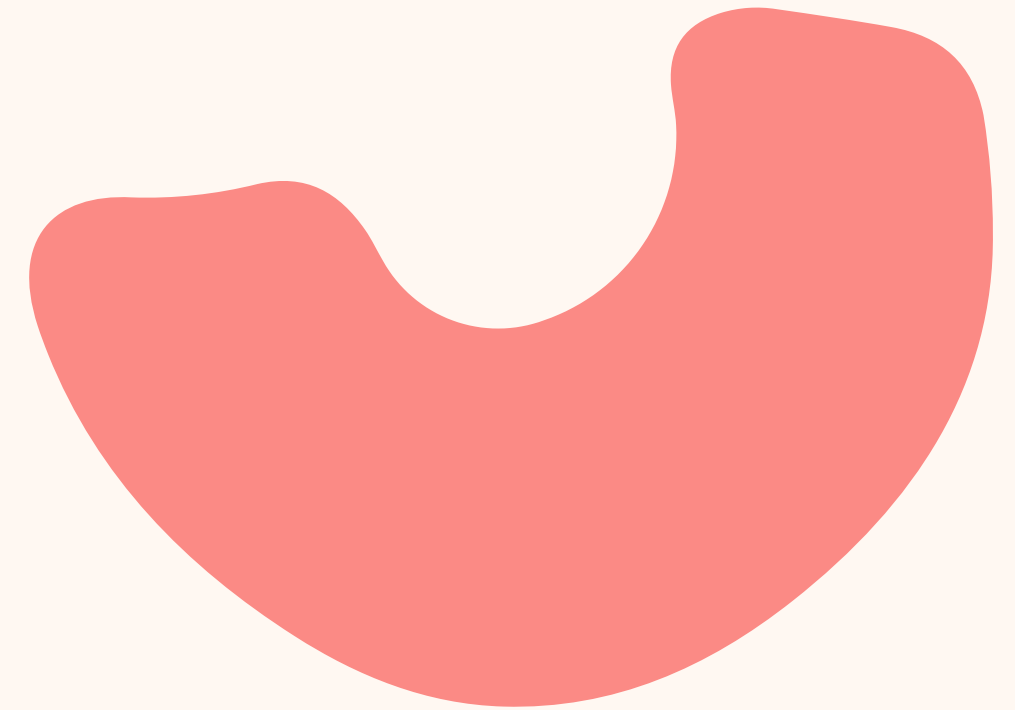
```
import csv
with open('marks.csv') as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```



DISPLAYING DATA ROW BY ROW

```
for row in reader:  
    print(row)  
    print(row[0]) # First column  
    print(row[1]) # Second column
```

Each row is a list! Access columns using index numbers starting from 0.



UNDERSTANDING THE OUTPUT

Output Example:

```
['Aarav', '85', 'Pass']
```

```
Aarav
```

```
85
```

Rows print as lists, and we can pick any column we want!

CODE-ALONG: READING THE STUDENT MARKS TABLE

Let's code together! Open "student_marks.csv" and read each student's data:

```
import csv
with open('student_marks.csv') as file:
    reader = csv.reader(file)
    for row in reader:
        print(row[0], row[1])
```

Output: Aarav 85, Priya 92, Rohan 78, Ananya 88



SCHOOL REPORT CHALLENGE

CSV has 5 students:

Name, Marks

Aarav, 85

Priya, 38

Rohan, 72

Your Task:

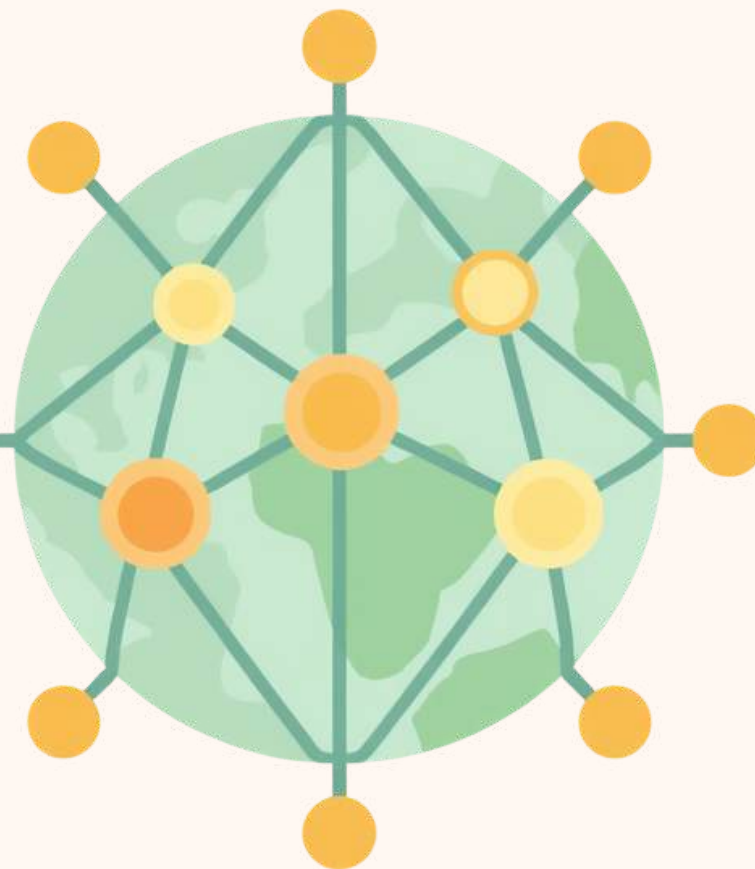
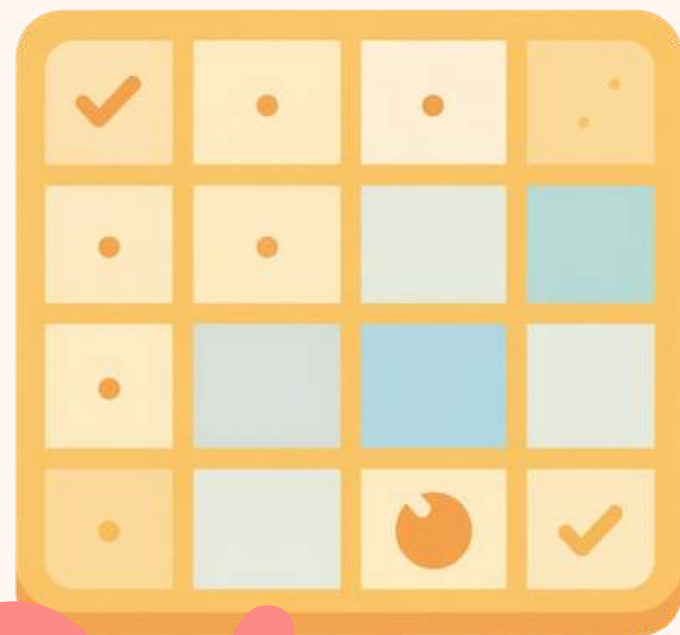
Print "Pass" if marks > 40

Print "Fail" if marks <= 40



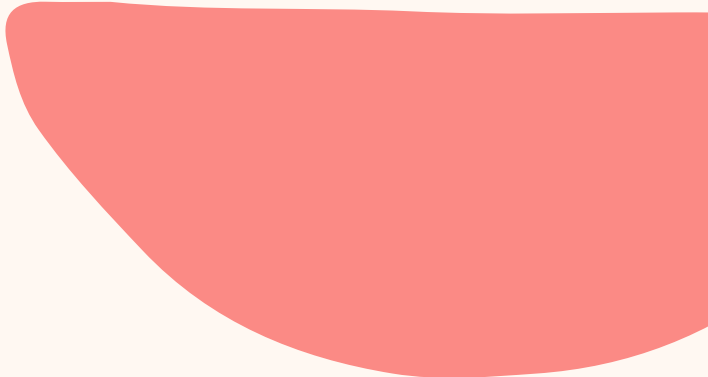
WHY USE CSV INSTEAD OF TEXT FILES?

CSV files are super helpful! Let's discover why programmers love using them to save and share data.





THE MAGIC OF
PERMANENCE:
CSV FILES
REMEMBER
FOREVER!



THANK YOU!

Thank you for learning about CSV with Python! Try reading CSV files at home. Next up: Writing data to CSV files!

